Machine Learning Component Brief description

The Machine Learning option that I chose was Event Recommender.

User Features (from interaction history)

Extracted in `build_user_feature_matrix` and `train_click_prediction_model` :

- **academic_interests** – count of academic events liked
- **social_interests** – count of social events liked
- **recreational_interests** – count of recreational events liked
- **wellbeing_interests** – count of wellbeing events liked
- **total_interests** – total liked events
- **interest_diversity** – number of categories the user liked
- **interest_ratio** – liked events / total events shown

Event Features

From `train_click_prediction_model` and `predict_event_click_likelihood` :

- **is_academic**
- **is_social**
- **is_recreational**
- **is_wellbeing**

These are combined into a single feature vector for Logistic Regression.

2. Model Choice (Plain Language)

- **KNN**: Finds students who behave similarly (e.g., similar event interests). If similar users liked an event you haven't seen, it recommends it.
- **Logistic Regression**: Learns patterns that predict whether a user will click on an event based on both user history and event type.
- **Hybrid**: KNN will recommend events to the user, while logistics Regression will prioritise them based on the user's top interest. Compared to utilising model alone, thus increases accuracy.

3. Assessment

The hybrid recommender in evaluate_event_recommendations, the script calculates precision@5, recall@5 and F1@5.

Examples:

- **Average prcision@5-** The percentage of events that were suggested
- **Average recall@5-** The percentage of events that were successfully suggested.
- **Average F1@5-**

This goes to show that the recommender is producing accurate results.